K9ESV · McHenry County Emergency Services Volunteers
and McHenry County Emergency Management Agency

FIELDCOMMS

Incident Management System v1.0

FIELD QUICK REFERENCE

Operator Quick Reference Card

RACES · ARES · Starcom · K9ESV · MCESV / MCEMA

July 2026

■ Getting Started & Dashboard

http://192.168.50.1/

STEP	WHAT TO DO
1	**Connect to EMCOMM-NET** — select it from your device's Wi-Fi list. No password required on the field network.
2	**Open any browser** — type **http://192.168.50.1** in the address bar and press Enter.
3	**Enter your identity** — type your callsign, Radio ID, or name when the prompt appears. Optionally add your ICS position. This is remembered by your browser.
4	**Select a dashboard mode** from the three buttons at the top of the page:

MODE	BUTTON	USE IT FOR
Amateur Radio	■	ARES/RACES nets, APRS, Winlink, JS8Call, NTS, callsign lookup, propagation
Starcom / Public Safety	■	Starcom nets, SAR nets, weather nets, resource tracking, unit roster
ICS	■	Incident command — all five ICS sections plus Planning P and ICS forms

Dashboard Elements

ELEMENT	WHAT IT DOES
Hero bar (top)	Station callsign, system name, live UTC clock and local 24-hr clock
NWS Weather Alerts	Live alerts color-coded by severity — click any alert to expand full details
Operation cards	Quick-launch tiles for every tool in the current mode — click to open
APRS station table	Live stations heard by Graywolf, with callsign, distance, and comment
Status sidebar	CPU/memory/disk/temp, service health dots (green=running), GPS, Dead Man's Switch state

■ The dashboard remembers your last-used mode between sessions. Return to Starcom mode directly from Starcom pages using the ■ Starcom Dashboard back link.

■ Amateur Net Control Logger

http://192.168.50.1/netcontrol.html

STEP	WHAT TO DO
1	**Open a net** — click **+ New Net**, enter the net name, frequency, mode, and type. Click **Open Net**.
2	**Log a check-in** — type a callsign and press **Enter**. FCC data fills automatically. Add location and remarks if needed.
3	**Log traffic** — click **+ Traffic**, enter message number, type, sender, addressee, and subject.
4	**Close the net** — click **Close Net**. The log is saved and available for export.
5	**Export ICS-309** — click **Export ICS-309** to generate the printable comms log.
6	**Share observer link** — click **■ Observer Link** to copy a read-only view URL for the EOC or served agencies.

FEATURE	HOW TO USE
Multiple nets	Each net has its own tab. Click a tab to switch between active nets.
FCC auto-fill	Type any US callsign — name and license class fill automatically from the offline FCC database.
Precedence levels	ROUTINE / WELFARE / PRIORITY / EMERGENCY — set per check-in or traffic entry.
Dead Man's Switch	Click ■ Arm on the Dead Man's Switch page to monitor this net for inactivity.

■ Starcom Net Logger

http://192.168.50.1/starcom.html

STEP	WHAT TO DO
1	**Open a net** — click **+ New Net**. Choose net type: Starcom General / Weather Net / SAR Net / Observer Net.
2	**Log a check-in** — enter **Radio ID** (unit number) and **Unit Name**. Press **Enter** or click **+ Check In**.
3	**Set unit status** — click the status badge on any check-in row to cycle: Available → Deployed → Staging → Out of Service.
4	**Run multiple nets simultaneously** — click **+ New Net** again for each additional net. Each open net appears as a badge in the active nets panel on the left. Click any badge to switch between nets. Each net keeps its own independent check-in log and traffic log.
5	**Close a net** — select the net, then click **Close Net**. All check-ins and traffic are archived.

FIELD	DESCRIPTION
Radio ID	Starcom unit number (e.g. 2301, 4710). Required for every check-in.
Unit Name	Agency and unit description (e.g. "McHenry SO — Unit 12")
Dispatch Ctr	Dispatching agency or center (e.g. MCECC, IDOT)
Weather / SAR	Use the quick-launch cards on the Starcom dashboard to pre-set net type
Multiple nets	Typical activation: Starcom General + Weather Net + SAR Net all open simultaneously

Member Roster

http://192.168.50.1/roster.html

TAB	WHAT TO DO THERE
Members	Add/edit members. Fields: callsign, name, Radio ID, phone, address, license class, email, notes.
Certifications	Check off ICS-100/200/300/400/700/800, EmComm I/II, CERT, First Aid, FEMA IS courses.
Equipment	Check off capabilities: HF/VHF/UHF radio, Winlink, JS8Call, APRS, generator, antenna, laptop, go-kit.
Activations	Log check-ins for this incident. Click **+ Check In** to mark a member as activated. Walk-ins can be added here.
Import/Export	Export roster to CSV. Import from a previous activation CSV to pre-populate for the next event.

Resource Board

http://192.168.50.1/resources.html

STEP	WHAT TO DO
1	**Add a resource** — click **+ Add Resource**. Enter name, type, owner/callsign, location, and initial status.
2	**Change status** — click the colored status badge on any card to cycle through the five states.
3	**Filter** — use the Type and Status dropdowns to show only the resources you need.

STATUS	COLOR	MEANING
Available	Green	On scene and ready for assignment
Assigned	Amber	Actively tasked
Staging	Blue	En route or at staging area
Out of Service	Red	Unavailable — hold, mechanical, or medical
Demobilized	Gray	Released and departed

Resource Tracking Map

http://192.168.50.1/resmap.html

ACTION	HOW
Place a unit	Click **+ Add Unit** → enter Radio ID, unit name, type, status → click ■ **Pick on Map** or enter lat/lon → Save.
Move a unit	Drag the unit marker to its new position on the map. Coordinates update automatically.
Change status	Click the marker → Edit → change Status dropdown → Save. Color updates immediately.
Draw a zone	Click **Draw Zone** → click map points to define boundary → double-click to finish → enter zone name.

■ ICS Platform — All Five Sections

http://192.168.50.1/ics/

Starting an Incident

STEP	WHAT TO DO
1	Go to **http://192.168.50.1/ics/** — or click the ICS Platform card from the ICS dashboard mode.
2	Click ■ **New Incident**. Fill in: Incident Name, Type, Location, Incident Commander, Op Period Duration.
3	Click **Activate Incident**. All five sections are now active with this incident loaded.
4	Navigate between sections using the tabs at the top: ■ Command · ■ Operations · ■ Planning · ■ Logistics · ■ Finance
5	Click ■ **Advance Period** at the start of each new operational period.

What Each Section Does

SECTION	KEY TOOLS
■ Command	Incident objectives (100-item dropdown + free text), safety message, weather summary, situation report, command staff roster. Forms: ICS-201, ICS-202, ICS-203.
■ Operations	T-card board — drag resources between Available / Assigned / Staging / Out of Service / Released. Click any card for assignment details. Forms: ICS-204.
■ Planning	IAP form tracker, resource status table, incident documentation log, operational period notes. Forms: ICS-209.
■ Logistics	Communications plan (ICS-205) with pre-filled McHenry County channels, supply tracking, meal log, personnel check-in list. Forms: ICS-205, ICS-206, ICS-211.
■ Finance/Admin	Expenditure log, personnel time tracking, procurement orders. Export to CSV for agency reporting.

> ■ The 100-item objectives dropdown in Command section is organized into 13 groups. Select an objective → it pre-fills the text box → edit if needed → click + Add. MCESV/MCEMA-specific objectives are in the last group.

■ ICS Forms Quick-Reference

http://192.168.50.1/

FORM	URL	WHAT IT DOES
ICS-213 General Message	/ics213.html	Formal written message. Fill sender, receiver, subject, message body. Print or save to incident.
ICS-214 Activity Log	/ics214.html	Unit activity log. Add personnel, then log timestamped activities. Print for IAP.
ICS-309 Comms Log	/ics309.html	Formal communications log. Add entries with timestamp, from, to, subject. Save to incident.
NTS Radiogram	/nts.html	ARRL formatted radiogram. Auto-fills date/time/number. All preamble fields. Print when complete.
Winlink Import	/winlink-import.html	Drag-and-drop ICS form XML from Winlink Express. Review fields, then archive to active incident.
Print Center	/printcenter.html	Print any blank ICS form, cheat sheet, or operator access card from one location. Printer options: A) own printer, B) USB printer shared via Pi/CUPS (admin: http://192.168.50.1:631), C) network printer on EMCOMM-NET.

■ Planning P — 15-Phase Cycle Guide

http://192.168.50.1/ics/planningp.html

Click the ■ **Planning P** tab on any ICS page. Click a phase button → see the standard agenda, required forms, and who should attend. Click **Generate briefing sheet** to print a one-page phase cover for the IAP.

COLOR	PHASES	GROUP
Gray	Phases 1–5	Initial Response — incident through incident brief
Yellow	Phases 6–7	Establish Objectives — IC/UC objectives + C&G; Staff Meeting
Red	Phases 8–11	Develop the Plan — tactics meeting through planning meeting
Green	Phases 12–13	Prepare & Disseminate — IAP prep + operations briefing
Teal	Phases 14–15	Execute, Evaluate & Revise — execute plan + new ops period

■ Tools Quick-Reference

http://192.168.50.1/

TOOL	URL	QUICK USE
FCC Callsign Lookup	/callsign.html	Type callsign → Enter. Results from local FCC database (~800K licensees). Click + Add to Roster.
APRS Tactical Map	/tactical.html	Live APRS stations from Graywolf/YAAC. Click station to center. Draw overlays with toolbar tools.
HF Propagation	/propagation.html	Live SFI, K-index, A-index from HamQSL. Band-by-band conditions. Useful for frequency selection.
Repeater Database	/repeaters.html	Filter by band, tone, mode, ARES affiliation. Click a row to copy frequency to clipboard.
Grid Square Calc	/grid.html	Enter lat/lon → get Maidenhead grid. Enter grid → get lat/lon + distance/bearing from you.
Pre-Flight Check	/preflight.html	GO / CAUTION / NO-GO readiness verdict. Run before every activation. Export report for records.
Dead Man's Switch	/deadmans.html	Click ■ Arm next to a net. Alerts you if the net goes silent beyond the configured threshold.
Observer Mode	/observer.html	Read-only live net view. Share the URL with EOC or served agencies — no identity prompt needed.
Reference Library	/refs.html	Upload PDFs/docs up to 200 MB. Tag, search, and share field documents across all devices.
Kiwix Library	:8081/	Offline Wikipedia, WikiMed, iFixit, Wikivoyage. No internet needed. Open from dashboard card.
Health Monitor	:5051/health	Live CPU/memory/disk/temp, all service status dots, GPS fix quality, internet status.
WAN Status	/wan-status.html	Active WAN source (InstyConnect cellular or Starlink satellite), signal strength, carrier, latency, connectivity tests, and WAN event log.
AMPRNet Gateway	192.168.50.2:9000	WireGuard tunnel state, AMPRNet 44.x.x.x address, traffic stats. Requires FCC callsign login. Tunnel control requires gateway Pi keyboard.
Facilities Directory	/facilities.html	EOC, shelters, hospitals, staging areas with address, phone, and radio frequencies.
Radio Cheat Sheets	/cheatsheets.html	Phonetic alphabet, Q-codes, prowords, band plan, CTCSS tones, ICS position titles.
JS8Call	dashboard card	Click the JS8Call card → enter Windows laptop IP when prompted → opens JS8Call web interface.

Winlink & Digital Comms

Windows laptop + IC-7300

METHOD	HOW TO USE
Winlink Express (primary)	Runs on Windows laptop. IC-7300 connected via single USB-A→USB-B cable. Use VARA HF or VARA FM mode. After sending/receiving an ICS form, save the RMS_Express_Form_*.xml and import it at /winlink-import.html.
Pat Winlink (backup)	Runs on the Pi at port 8090. Open from dashboard card. Browser-based. Use when Windows laptop is unavailable.
JS8Call (digital messaging)	Runs on Windows laptop. Enable API at File → Settings → API → port 2442. Click the JS8Call dashboard card → enter Windows laptop IP. Recommended frequency: 7.078 MHz USB-D (40m calling).

Network — Admin URLs & IP Reference

All addresses on EMCOMM-NET (192.168.50.0/24)

DEVICE / SERVICE	ADDRESS	NOTES
FieldComms dashboard	http://192.168.50.1	Main entry point — all 32 tools
WAN Status dashboard	/wan-status.html	InstyConnect + Starlink + WAN event log
AMPRNet gateway status	http://192.168.50.2:9000	Requires FCC callsign login
ASUS router admin (primary)	http://192.168.50.254	Wi-Fi settings, dual WAN failover, AiMesh
InstyConnect modem admin	http://10.1.1.1 or http://my.insty	Connect to InstyConnect Wi-Fi first
Starlink dish admin	http://192.168.100.1	Or use Starlink app on phone
CUPS print server	http://192.168.50.1:631	Printer management and shared queue
Pat Winlink	http://192.168.50.1:8090	Browser-based Winlink backup
Kiwix offline library	http://192.168.50.1:8081	Wikipedia, WikiMed, iFixit offline
Health monitor API	http://192.168.50.1:5051/health	JSON health data for all services
44Net tunnel control	http://localhost:9001	Gateway Pi keyboard only — callsign login required

WAN Connectivity Quick Reference

InstyConnect Primary · Starlink Automatic Failover

WAN SOURCE	HOW CONNECTED	WHEN ACTIVE
InstyConnect Drum	PoE cable from outdoor antenna to ASUS WAN port	Default — primary WAN at all activations · 10.1.1.1
InstyConnect Switchblade	Same PoE cable to same ASUS WAN port (swap Drum)	When Drum signal is poor — aim toward tower · 10.1.1.1
Starlink satellite	Ethernet adapter + USB adapter → ASUS USB WAN	Auto when cellular drops — switches in 60s · 192.168.100.1
Site Ethernet	Site cable to ASUS WAN port	Manual — when site internet available · 192.168.50.254

EMCOMM-NET Coverage — AiMesh Nodes

Three ASUS RT-BE58 Go routers — same SSID, seamless roaming

ROUTER	SWITCH PORT	PLACEMENT
ASUS RT-BE58 Go (primary)	Uplink — Port 1	Central position — command post or EOC entrance. Manages WAN and DHCP.

ROUTER	SWITCH PORT	PLACEMENT
ASUS RT-BE58 Go (mesh node 1)	Port 11	Secondary room, opposite wing, or upper floor.
ASUS RT-BE58 Go (mesh node 2)	Port 12	Third coverage zone — outdoor staging, parking, or far wing.

> ■ All three routers broadcast EMCOMM-NET on 2.4 GHz and 5 GHz. Devices roam between them automatically — no reconnection or password re-entry needed. Pairing a new mesh node to the primary takes under 5 minutes via http://192.168.50.254 → AiMesh → Add Node.

Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
Cards give errors	Not on EMCOMM-NET — device is on a different network	Check Wi-Fi — must show EMCOMM-NET. Reconnect if needed.
Cannot reach http://192.168.50.1	Pi not running or ASUS router not powered	Check Pi power (green LED). Check ASUS router power. Wait 60s after Pi boot.
FCC lookup no results	FCC database not built or not a US amateur callsign	Run: sudo systemctl start fcc-refresh.service (needs internet)
APRS map no stations	Graywolf/YAAC not running or no RF received yet	Check Health Monitor for service status. Confirm antenna connected to IC-7300.
Winlink import fails	Wrong file — must be the XML attachment, not message body	In Winlink Express, right-click the .xml attachment → Save As → then import that file.
Service dot is red	A background service has stopped	SSH to Pi: sudo systemctl restart (e.g. fcc-lookup, ics-platform)
WAN shows offline	InstyConnect disconnected or plan inactive	Check Drum LED. Try Switchblade. Check plan at instyconnect.com. If Starlink is connected, failover should be automatic.
Starlink failover not working	USB adapter not seated or ASUS Dual WAN not enabled	Check ASUS admin → WAN → Dual WAN → must be ON with USB as secondary.
AMPRNet login fails	Callsign not in FCC DB or FieldComms Pi unreachable	Verify callsign is valid at http://192.168.50.1/callsign.html. If FCC DB offline, format-only validation is used as fallback.
Cannot reach 192.168.50.2:9000	Gateway Pi off or not on EMCOMM-NET	Check gateway Pi power (green LED). It must be connected to UniFi switch.

For any issue not covered here — open the Health Monitor at http://192.168.50.1:5051/health or check the system log with: journalctl -u fcc-lookup -n 50

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RACES · ARES · Starcom
http://192.168.50.1

FieldComms IMS v1.0 July 2026

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	**Cortex-A76 @ 2.4 GHz (Pi 5 generation)**	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	**8 GB LPDDR4X**	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	**TBD (current model)**	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	**✓ YES — Pi 5 chip, 8 GB RAM, current generation**	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

■ The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit ($120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit ($120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

■ **501(c)(3) Grant Funding Note:** MCESV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	**Pi 500 + Pi Monitor**	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed

■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.